Semantic Ontologies in Economics

Special Issue Proposal

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Abstract

A computed model's meaning is largely implicit in prose, notation, calibration, and code. Professional convention has helped fill the gaps. But economists increasingly use large language models (AIs) to write, modify, and translate modeling code, and AI output is probabilistic by construction. Unverified, such code can compute a different model from the one intended, its results cannot be cross-checked or reused, and its economic interpretation becomes ambiguous. Verification requires a semantic ontology: an AI-readable, deterministic statement, independent of code and solver, of which economic objects exist, what stands for what, what is written down, and when the interpretation holds. We propose a special issue collecting semantic ontologies for modeling languages, toolkits, empirical methods, and model classes.

1 Introduction and Proposal

When economists compute a model, its precise *meaning* is distributed, much of it implicitly, across prose, notation, calibration, and code. The paper says "we solve the following model", but its equations underdetermine both what its code computes and what the computations mean, leaving the reader to fill the gaps. The same gaps confront whoever rebuilds or extends the model. As a result, without a concrete statement of the model's meaning to check against, computational results are hard to cross-verify, and interoperability between implementations becomes difficult.

So far, professional convention has helped trained modelers within a discipline to interpret, replicate, and build on one another's models. But convention breaks down once AI is used to write, modify, and translate modeling code. To interpret or verify AI output we need a statement, independent of the code, of which economic objects are computed and which relations among them are enforced; we call such statements *semantic ontologies*. Without semantic ontologies, AI use is prone to error and misunderstanding. With semantic ontologies, translating models between toolkits, and even generating new modeling research, become operations a language model can carry out and be checked on cheaply, rather than manual recoding.[1]

> **Proposed special issue**
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> We propose a special issue that collects semantic ontologies for the domains of computational economics: modeling languages, toolkits, empirical methods, and model classes. Each paper takes one domain of study, states its semantic ontology, and demonstrates it on worked examples of the team's own choosing. The SCE working group will submit a subset of the papers, an open call will invite the rest, and a comparison paper, written jointly by the participating teams, will close the issue.

1.1 Semantic Ontologies

Semantic ontologies are broadly defined as a formal way to codify the meaning of computational models. For our purposes, a semantic ontology consists of the objects and relations within the model, the meaning of those objects and relations, and the written forms that record them (syntax: a file, a model write-up, a specification).[2] Theory supplies part of the semantic ontology (a general-equilibrium model's objects are precisely defined), but the semantic ontology must collect these definitions, map them concretely to computational and written counterparts, and state the assumptions under which the mapping holds. Without a written semantic ontology, nothing says which of the theory's objects a given file or function call stands for, and ordinary solver code gives one executable realization, not a solver-independent statement of what the representation denotes.

The importance of semantic ontologies is broadly recognized across research fields and in industry. Manufacturing's Process Specification Language (Grüninger and Menzel 2003; Bock and Grüninger 2005), Modelica for physical systems (Fritzson and Engelson 1998; Modelica Association 2023), planning's PDDL2.1 (Fox and Long 2003), and neuroscience's NeuroML (Gleeson et al. 2010) each attach an explicit, solver-independent meaning to a model representation. Industry has met the same need from the opposite direction, reconstructing never-written conceptual models of the business from its data and business processes (AWS Database Blog 2026). Compared with all of these, economics is well placed: the relations among a model's objects are the theory itself, already stated precisely in the

[1]The nearest precedent in economics, the Macroeconomic Model Data Base (Wieland et al. 2012), compares models under common variables, common shocks, and a menu of common policy rules while each model keeps its own equations; it standardizes comparison, not meaning.

[2]In a heterogeneous-agent model, for instance, the ontology contains the response of decision rules to prices, the cross-sectional distribution decisions induce, and the feedback of that distribution into prices, both as relations of the theory and as their computational counterparts.

paper. What remains unstated is the map from the representations economists compute with (model files, toolkit calls, estimation specifications) to that theory, and the conditions under which the map holds. Stating these maps, domain by domain, is the work this issue proposes.

2 Guidelines for Papers

2.1 Research Domains and Themes

Domains of study. Each paper in the special issue will address one *domain of study*. Domains of study can include a modeling language, a toolkit, an empirical method, a model class, or a combination of these.[3]

DOMAINS OF STUDY

- **Modeling languages.** A modeling language is a fixed grammar in which a complete model is written as a file, as in Dynare's model language or Dolo's YAML model files.
- **Toolkits.** A toolkit is a collection of construction calls and classes from which a model is assembled in a programming language, as in HARK's agent classes, the sequence-space Jacobian toolkit (Auclert et al. 2021), or the VFI Toolkit.
- **Estimation and empirical methods.** An empirical method takes a model to data. It states which measured objects stand for the model's quantities (for example, prices and shocks), which classifications organize those objects, and which transformations construct the inputs to estimation or calibration. The simulated method of moments and indirect inference are examples of empirical methods, and so is the calibration of a computable general-equilibrium model to a social accounting matrix or input-output table.
- **Model classes.** A model class is a family of models built from the same kinds of objects, as in heterogeneous-agent macroeconomies or overlapping-generations economies.

Research themes. The research themes of the issue are two of the core research questions each paper will answer: first, constructing a semantic ontology for its domain of study, and second, establishing the ontology's *metatheory*, statements about the ontology rather than components of it. There are three components to each semantic ontology: what is assumed to exist (the *ontology*), what stands for and relates to what (the *denotation*), and

[3]Whichever domain a paper addresses, its theory and mathematics are taken as pre-existing: the Bellman equation is not in question, while what a given Dynare file or HARK model means is. A language, toolkit, or empirical-method paper therefore emphasizes the map from representations to the mathematical objects they stand for, while a model-class paper emphasizes the objects and relations implied by its mathematical and theoretical framework. Differences among the stated ontologies are results for the closing comparison to report.

what is written down (the *syntax*). Domains of study will differ in how important these components are, and some, such as a model class, may not have certain components, such as syntax. The metatheory states when the interpretation holds (*well-posedness*) and what preserves it (*equivalence*, *adequacy*, and *convergence*). Details of these research themes are given in Appendix A.

2.2 Methodology

Formalizing the semantic ontology. A requirement for each paper will be that it presents a *formal* account of the semantic ontology of its domain of study. However, the researchers choose the formalism in which they state these components.

The available formalisms differ in how much meaning they fix, and they can be ordered from least to most (Uschold and Grüninger 1996).

At one end, a team records which entities exist and which relations connect them, as boxes and arrows: an entity-relationship diagram (Chen 1976), a UML class diagram (Berardi, Calvanese and De Giacomo 2005), or a knowledge graph (Hogan et al. 2021). These *graphs and diagrams* name the model's objects but place few restrictions on what they mean.

Writing the ontology as *logical axioms* fixes more: the axioms rule out interpretations of the objects much as parameter restrictions rule out models. The axioms can be stated in any formal logic. In the older tradition this meant first-order logic (Gruber 1993), and in much current practice it means a description logic such as the Semantic Web's Web Ontology Language (OWL) (Baader et al. 2017; W3C OWL Working Group 2012).

The most is fixed by the *mathematical semantics* developed for programming languages, which attach an explicit mathematical meaning to every written form. Denotational semantics assigns each form an object in the ordered structures of domain theory (Scott and Strachey 1971). Initial-algebra semantics takes meaning to be the unique homomorphism from a many-sorted algebra of terms (Goguen, Thatcher, Wagner and Wright 1977). Typed categories interpret a typed syntax in a category with matching structure (Lambek and Scott 1986). Operational semantics gives rules for execution on an abstract machine (Plotkin 1981), and axiomatic semantics gives the assertions that hold before and after execution (Hoare 1969).

Practice differs by field. Google and other technology firms record knowledge graphs from observed data, without axioms (Noy et al. 2019). The Gene Ontology Consortium states its ontology of gene functions in description logic (Ashburner et al. 2000). A few programming languages, such as Standard ML and WebAssembly, have been given complete formal semantics (Milner, Tofte, Harper and MacQueen 1997; Haas et al. 2017), though most languages in use have not.

How far along this scale a paper goes depends on whether its domain of study has a syntax. A team whose domain has one (a modeling language or a toolkit) can use any of the mathematical semantics; a team whose ontology has no syntax (a model class, or an empirical method before its specification is constructed) can axiomatize the ontology in a logic or record it as a graph. Declarative languages make the mathematical route easiest: a Dynare or Dolo file states the model itself rather than a procedure for solving it, so the file is already the kind of written form a semantic map can interpret. A team seeking machine-checked guarantees can go further and formalize its domain in a proof assistant such as Lean, defining the model objects as types and the file-to-object map as a function inside the system, so that well-posedness and equivalence become theorems the machine verifies (de Moura and Ullrich 2021).

Top-down vs. bottom-up approaches. A team can also build its semantic ontology top-down or bottom-up. Building top-down, it starts from the theory and formalizes the theory's objects; building bottom-up, it starts from observed data and relations and abstracts the ontology from them, as industry does when it reconstructs a conceptual model from a data catalog (AWS Database Blog 2026). In this issue we expect economic theory to supply the structure, so most papers will work top-down; a bottom-up construction may suit agent-based modeling or policy research, where observed relations precede a settled theory.

3 Organization

Papers pass an internal review within the working group [process TBC] and then the journal's ordinary external refereeing. We expect six to eight papers, with a session at the Society's conference between submission and revision.

3.1 Teams and Editors

The working group's members include developers of Dynare, HARK, and the VFI Toolkit. Committed teams: [to be listed; only teams that have agreed in writing]. The working group's papers will come from its subgroups. Akshay Shanker and Matthew McKay chair the working group and coordinate the teams and the internal review. The guest editors (a lead editor and at least one co-editor, neither submitting to the issue) are drawn from outside the working group [names, institutions; TBC]. Papers are submitted through the journal's editorial system, marked for this special issue, and are refereed under the journal's ordinary standards; the guest editors handle every paper, including the comparison paper, subject to the journal's final editorial authority.

3.2 Timeline

Calendar dates TBC on acceptance.

- Call for papers [on acceptance]
- Submissions [+9 months]

- Internal review completed [+12 months; process TBC]
- Referee reports [+15 months]
- Revisions and the comparison paper [+24 months]

A Research Themes

A paper addresses those of the following five research themes relevant to its domain of study. The first three are the components of the domain's semantic ontology; the last two are statements about it (when the interpretation holds, and what preserves it under changes of representation and computation):

THE RESEARCH THEMES

1. **Ontology: what is assumed to exist.** The economic and mathematical entities and relations within the domain (agents, states, shocks, timing, the equilibrium concept, operators, functions, and which objects determine which), together with the criteria for when two of them are the same (Gruber 1993; Guarino, Oberle and Staab 2009). The ontology is stated in its own terms, without reference to the syntax.

2. **Denotation: what stands for what.** The meaning of the ontology's objects. The meaning takes the form of relationships between the mathematical, economic and computational representations, for instance value functions as economic concepts, instantiated on a mathematical space, and approximated on the computer. Typically the denotations will also include relationships between these representations and the written forms: each written form stands for its mathematical or economic object under an explicit map (Harel and Rumpe 2004).

3. **Syntax: what is written down.** The set of legal written forms: the model file, the sequence of construction calls, or the specification of an estimation exercise. For instance, a language has a full grammar; a toolkit may expose construction calls or classes. A domain (such as an empirical method) may have no syntactic written form at all, in which case much of the role of syntax is played by the denotation.

The next two research themes are not components of the semantic ontology; they are statements about it:

4. **Well-posedness: when the interpretation holds.** The conditions under which the denotation is well defined: domains, units, timing, information structure, and the parameter restrictions assumed.

5. **Equivalence, adequacy, convergence: what preserves it.** Denotational equivalence: two written forms, one denotation. Adequacy: an implementation computes exactly the object denoted. Convergence: a numerical approximation approaches it as grids and tolerances are refined.

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